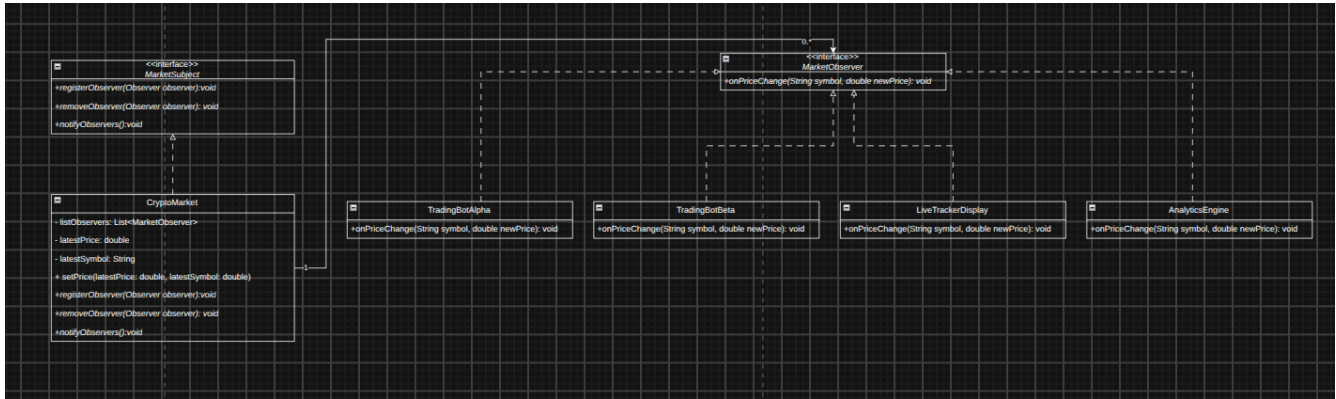


Lab 3 Independent Exercise

UML Class Diagram



Code outputs

Base test:

```
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 59500.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 72000.0
```

Removing Alpha:

```
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 59500.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 72000.0
Removing Trading Bot Alpha...
[Live Tracker Display]: Symbol is: BTC Price is: 4999.0
```

Adding AnalyticsEngine:

```

[Analytics Engine] Average observed price is: 59500.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 59500.0
[Analytics Engine] Average observed price is: 65750.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 72000.0
Removing Trading Bot Alpha...
[Analytics Engine] Average observed price is: 45499.666666666664
[Live Tracker Display]: Symbol is: BTC Price is: 4999.0

```

C

Adding additional three price updates:

```

[Analytics Engine] Average observed price is: 59500.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 59500.0
[Analytics Engine] Average observed price is: 65750.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 72000.0
Removing Trading Bot Alpha...
[Analytics Engine] Average observed price is: 45499.666666666664
[Live Tracker Display]: Symbol is: BTC Price is: 4999.0
[Analytics Engine] Average observed price is: 36374.75
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: Gold Price is: 9000.0
[Analytics Engine] Average observed price is: 32099.8
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 15000.0
[Analytics Engine] Average observed price is: 27083.166666666668
[Live Tracker Display]: Symbol is: Silver Price is: 2000.0

```

Adding the EmailPriceAlert:

```

[Analytics Engine] Average observed price is: 59500.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 59500.0
[Analytics Engine] Average observed price is: 65750.0
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 72000.0
Removing Trading Bot Alpha...
[Analytics Engine] Average observed price is: 45499.666666666664
[Live Tracker Display]: Symbol is: BTC Price is: 4999.0
[Analytics Engine] Average observed price is: 36374.75
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: Gold Price is: 9000.0
[Analytics Engine] Average observed price is: 32099.8
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: BTC Price is: 15000.0
[Analytics Engine] Average observed price is: 27083.166666666668
[Live Tracker Display]: Symbol is: Silver Price is: 2000.0
[Email Price Alert] Sending Email...
[Analytics Engine] Average observed price is: 31785.571428571428
[Trading Bot Beta]]: Beta Bot executing SELL order!
[Live Tracker Display]: Symbol is: ETH Price is: 60000.0

```

Independent Questions

1. Which class is the Concrete Subject, and which classes are Concrete Observers?

Concrete Subject is the CryptoMarket

Concrete Observers are LiveTrackerDisplay, AnalyticsEngine, TradingBotAlpha, TradingBotBeta, and EmailPriceAlert classes

2. Why can each trading bot apply a different rule without adding bot-specific conditions to CryptoMarket?

CryptoMarket is just sending the latest data, observers take that data and apply what they want to it, which may (and probably) be different among observers.

3. What existing code must change when EmailPriceAlert is added?

No code need to change, just extend the MarketObserver interface and then you can get the data once registered.